# 5.3 Interpreting Equations

## Lesson Introduction

| | | | |
|---|---|---|---|
| Benchmarks | MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.<br><br>MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.<br><br>MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options. | Learning Targets | • I can define variables in a real-world context.<br>• I can interpret equations and their parts in a given context. |

| Benchmark Coherence | Building On | Working Toward |
|---|---|---|
| | MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides. | N/A |

| | |
|---|---|
| Essential Questions | • How do I define variables in a real-world context?<br>• How can equations and their parts be interpreted in a given context? |
| Lesson Narrative | In this lesson, students will be introduced to the concept of a parameter. They will use this knowledge to help define variables in real-world contexts and interpret equations in a given context. Students will encounter several real-world problems throughout this lesson. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 5.3.1 Warm-Up (~5 min.) | Orange juice activity *(SE p. 240)* | 5.3.3 Activity (20-25 min.) | The meaning of values card sort activity *(SE p. 243)* |
| | 5.3.2 Guided Instruction (10-15 min.) | Introduction to parameters *(SE p. 241)* | 5.3.4 Wrap-Up (~5 Min) | Growth Mindset M.I.S.T.A.K.E.S activity *(SE p. 243)* |

| Resources | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| | <u>Key Terms:</u><br>• Parameter<br><br><u>Additional Terms:</u> N/A | • Card Sort Activity Cards | • 5.3.3 Card Sort activity will need to have materials prepared prior to launching this lesson. |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may misinterpret the y-intercept as the actual beginning year in questions that talk about “years since.”<br>• Students may reverse the interpretation of the slope in a given context.</td><td>• Students will engage in real-world problems throughout this entire lesson. Consider leveraging your existing word problem strategies throughout the lesson to support students as they engage in word problems.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Three Reads<br>This lesson provides students the opportunity to collaborate on real-world problems. For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.</td><td>MA.K12.MTR.7.1: Real-World Contexts<br>This lesson consists solely of real-world problems. During this lesson students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Text-heavy lessons like Lesson 3 require students to apply skills of reading, decoding, and explaining in addition to the interpretation of the mathematics in contexts. Consider partnering with an English teacher as a joint lesson in decoding, or partner with an English teacher in providing scaffolding for struggling readers.<br>• Three Reads routine<br>• Partner Reading strategies<br>• Close Reading techniques</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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# 5.3.1 Warm-Up: Orange Juice

***Component Overview:*** *Students will answer three questions related to the nutritional values of orange juice.*

## 5.3.1 Warm-Up: Orange Juice

1. A typical orange produces $86$ grams of juice. The juice contains $9$ grams of carbohydrates.

   a. If $1$ gram (gm) of carbohydrate contains $4$ calories, how many calories come from the carbohydrates in one orange?
   36 *calories*

   b. The total number of calories of the juice from one orange is $39$. What percentage of those calories comes from the carbohydrates in the juice?
   92.3%

   c. Carbohydrates include sugar. The sugar that occurs in orange juice is natural sugar. If a typical orange contains $7$ gm of sugar, what percentage of the calories from carbohydrates comes from sugar? Round to the nearest tenth.
   $\frac{28}{36} = 0.\overline{7} \approx 77.8\%$

*Consider bringing orange juice to the classroom for students to engage in the Warm-Up with a tasty treat – increase engagement as often as possible.*

# 5.3.2 Guided Instruction: What Is a Parameter?

***Component Overview:*** *Students will be introduced to the concept of a parameter through one mathematical context and two real-world contexts.*

### 5.3.2 Guided Instruction: What is a Parameter?

1. In the expression $4x$, the $4$ is the coefficient and the $x$ is the variable. The expression $4x$ could be used to represent the number of calories in orange juice that comes from carbohydrates since $1$ gram of carbohydrate contains $4$ calories.

   A **parameter** is a quantity that influences the output.

   In the expression $4x$, what is the parameter?
   *$4$ is the parameter.*

*Consider creating a visual aid, such as an anchor chart, that shows a side-by-side comparison of a real-world and mathematical context. The visual aid should showcase the parameter of each so that students have a reference point.*

2. The cost of an item is $\$8.50$.

   a. Write an expression that can be used to determine the cost of $x$ items.
   $8.5x$

   b. Discuss with your partner why $\$8.50$ is a parameter. Summarize your discussion.
   *Answers will vary. It is a parameter because the output is determined by the input. The total cost will be dependent upon the total number of items.*

3. The price of an item is $P$. During a sale, the price is discounted by $20\%$, so the new price is $0.8P$. This could also be expressed as $(1 - 0.20)P$.

   Complete the table.

| Expression | Parameter |
|---|---|
| $0.8P$ | $0.8$ |
| $(1 - 0.20)P$ | $(1 - 0.20)$ |

*How can you determine the parameter of each of these expressions?*

# 5.3.2 Guided Instruction: What Is a Parameter? (Continued.)

4. Discuss with your partner which values in the U-Pick Blueberries sign are parameters. Justify your choice.
   *$4$ is the parameter because the total cost is dependent upon the number of pounds. The parking cost is fixed and would not be impacted by an input.*

**U-Pick Blueberries**

**\$4 per pound**

**\$1 parking**

*After students have an opportunity to complete question 4 with their partner, allow them to share out their responses in a whole-group discussion environment.*

5. The average price, $P$, in dollars, of a $16$ ounce bag of potato chips for each year, $x$, since 1980, can be modeled by the function $P(x) = 0.07x + 2.20$.

   a. Which value in the function is a parameter?
   $0.07$

   b. What does the value $2.20$ mean in terms of the price of potato chips?
   *The starting or initial cost of a bag of potato chips in $1980$.*

   c. What does the value $0.07$ mean in terms of the price of potato chips?
   *The rate or amount the price of a bag of potato chips increases each year since $1980$.*

*How is the parameter and rate of change related?*

6. The students in the Visual and Performing Arts program at Booker High School in Sarasota County, Florida are planning a field trip to the Ringling Art Museum. The cost of admission is $\$5$ and the cost of renting a bus is $\$115$. There will be $9$ chaperones. The budget for the field trip is $\$600$.

   a. Work with your partner to determine which values, if any, are parameters. Write a justification for each.

| Value | Parameter? | Justification |
|---|---|---|
| 5 | ○ yes<br>○ no | *The total cost will be based on the product of 5 and the input or number of tickets.* |
| 9 | ○ yes<br>○ no | *The number of chaperones is fixed.* |
| 115 | ○ yes<br>○ no | *The cost to rent a bus is fixed.* |
| 600 | ○ yes<br>○ no | *The amount budgeted is fixed.* |

   b. Use the values to complete the equation that can be used to find the number of students who can go on the field trip, $x$.

$$5(x+9)+115=600$$

*Three Reads*

*The Three Reads language routine provides decoding and comprehension strategies for text-heavy contexts. Consider modeling a Three Reads for question 6 to give students an example of how to interpret parameters in the Card Sort.*

- *First Read: Initial reading to orient to the context and figure out what is happening*
- *Second Read: Closer reading to specifically determine what the question is asking*
- *Third Read: This reading identifies important information using Mark-Up text that may contribute to answering the question.*

# 5.3.3 Activity: Card Sort

***Component Overview:*** *Students will complete a card sort for three different real-world contexts where they will match the meaning of each value and the equation that represents the scenario.*

## 5.3.3 Activity: Card Sort

1. Work with your partner to sort the cards provided by your teacher.
   a. Record the two lowercase letters on the card that matches the meaning of each value.

| Real-World Context | Meaning of Each Value | |
|---|---|---|
| 1 | 1991 | kk |
| | 6.88 | aa |
| | 0.37 | rr |
| | 3 | ww |
| 2 | 2010 | bb |
| | 19.116 | mm |
| | 1.352 | tt |
| | 43.452 | hh |
| 3 | 8 | yy |
| | 0.28 | gg |
| | 34.60 | nn |
| | 2.40 | dd |

   b. Record the letter and the equation that matches the context for each scenario.

| Real-World Context | Equation |
|---|---|
| 1 | B |
| 2 | L |
| 3 | W |

*For this activity, students will need to have access to the real-world context cards. Consider printing them on different colored paper to keep the sets of cards together in between classes if the activity will be done multiple times.*

# 5.3.4 Wrap-Up: Growth Mindset

***Component Overview:*** *Students will identify their favorite mistake from this lesson.*

**5.3.4 Wrap-Up: Growth Mindset**

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

   My favorite mistake from today was ...
   *Answers will vary.*

**Means**
**I**
**Start**
**To**
**Acquire**
**Knowledge**
**Experience**
**Skills**

*Consider creative alternatives for this wrap-up.*

- *Students can choose to explain their favorite mistake to a friend or family member, submitting their scratch work they used to illustrate their mistake.*
- *Students can post their work on social media and see which of their followers (or friends) can identify their mistake.*